Zulkefal

Blockchain & Backend Developer

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Portfolio: https://zulkefal.github.io/

Professional Summary

Blockchain and backend developer building secure, gas-optimized, production-grade systems for EVM networks. Experience spans cross-chain payment protocols, multi-token fee routing and stablecoin swaps, oracle and dispute-resolution systems, governance tokens, and upgradeable architectures, with mainnet deployments across EVM chains and two protocols audited by Hacken. Comfortable on both sides of the stack: Solidity contracts and Foundry test suites, signature-verification flows, and Node.js/NestJS services with ethers.js, The Graph indexing and scheduled jobs. Active contributor to Cyfrin CodeHawks audit competitions.

Experience

Blockchain & Backend Developer Sep 2024 – Present |Islamabad

Quecko Inc

- Built and maintained upgradeable smart contracts using UUPS, Transparent Proxy, Beacon Proxy and Diamond Proxy (EIP-2535), ensuring secure upgrade flows and storage-layout compatibility.
- Deployed contracts across multiple EVM-compatible blockchains.
- Deployed a protocol to production that currently holds **$97.5** in total value locked (TVL).
- Integrated Uniswap for on-chain token swapping and liquidity operations.
- Implemented ECDSA signatures and Merkle proofs for secure user whitelisting and validation.
- Built cross-chain functionality to bridge tokens between different blockchains.
- Developed backend services in Node.js and NestJS (TypeScript) that connect applications to the deployed smart contracts.
- Built NestJS APIs that generate and serve the ECDSA signatures and Merkle proofs used for whitelisting and claims.
- Used ethers.js in backend services to read on-chain state, send transactions and listen for contract events across multiple EVM chains.
- Built and deployed subgraphs on The Graph to index contract events and serve on-chain data to applications through GraphQL.
- Implemented scheduled cron jobs in Node.js to monitor on-chain state and run recurring backend tasks.

Projects

Tomi Pay — Cross-Chain Payment & Tokenomics Protocol Solidity, Foundry, Wormhole

- Architected a multi-chain payment protocol in Solidity that accepts any ERC-20 token, deducts configurable fees, and routes payments across BSC, Arbitrum, Avalanche, and Celo. Contract Link
- Integrated Wormhole cross-chain messaging to bridge tokens and trigger threshold-based burn/mint actions, coordinating a deflationary tokenomics model across multiple blockchains via dedicated sender/receiver contracts.
- Automated DeFi swap routing through Uniswap V2/V3 Universal Router with Permit2 approval delegation, converting heterogeneous fee tokens into USDT and the governance token (TOMI) in a single batched transaction.
- Implemented ECDSA signature-based token claiming with deadline validation and replay protection (signature tracking) for secure referral/incentive distribution.
- Developed a Foundry test suite with mainnet-fork testing, validating swap mechanics, signature flows, and cross-chain delivery against live chain state; configured a multi-chain deployment pipeline with per-chain RPC config and Etherscan verification.

Audited by Hacken - Decentralized Oracle & Dispute Resolution Protocol

- Designed and implemented a fully on-chain oracle voting protocol with time-bound questions and quorum-based finalization.
- Built three core modules:
  - **Survey:** Users create surveys and fund rewards; participants submit responses and receive payouts trustlessly onchain. Contract Link
  - **Question Resolution Module:** Users create questions finalized via oracle voting, requiring a majority quorum (N/2 + 1) with on-chain time extension logic if quorum is not met. Contract Link
  - **Dispute Resolution Module:** Enabled users to create disputes, support multi-party participation, submit proofs, and resolve cases via oracle-based adjudication. Contract Link
- Implemented multi-token payment support, allowing users to pay protocol fees in various ERC20 tokens.
- Integrated Uniswap V3 for on-chain token-to-ETH conversion to fund automated time extension functions.
- Architected contracts using UUPS upgradeable proxies.
- Implemented ECDSA-based signature voting.
- Applied gas optimization techniques across storage layout, voting logic, and fee handling.
- Wrote comprehensive Foundry test suites covering core logic, edge cases, and upgrade safety.
- Deployed and maintained the protocol on EVM-compatible mainnets.

13/06/2025 – 25/06/2025

WormHole Token Bridge (Cross-Chain Token Bridging) **GitHub**

- Developed a cross-chain token bridge using Solidity and the Wormhole protocol to transfer tokens between blockchains.
- Implemented smart contracts for message relaying and token distribution.
- Utilized Foundry for contract testing, deployment scripting, and environment setup with RPC endpoints.
- Solely responsible for architecture, smart contract development, and deployment across multiple blockchains.

01/08/2025 – 15/08/2025

Audited by Hacken — Governance Token Smart Contract

- Designed and implemented an ERC-20–based governance token with burnable and mintable functionality.
- Added a custom tokenomics mechanism to allow minting of 10% of burned tokens after a 24-hour cooldown period.
- Ensured secure access control and safeguards around minting logic to prevent abuse and unauthorized inflation.
- Developed comprehensive Foundry test cases covering minting, burning, time-based constraints, and edge cases to ensure correctness and attack resistance.

01/02/2023 – 15/06/2024

Token Presale & Vesting Smart Contract

- Built a token presale contract with partial token distribution at TGE and vesting for remaining allocations.
- Implemented a referral rewards mechanism to incentivize user participation.
- Added secure claim logic and time-based vesting, validated with comprehensive Hardhat tests.

Volunteering

02/08/2024 – CURRENT

Smart Contract Audits (Cyfrin CodeHawks Competitions)

Actively participate in multiple smart contract audit contests hosted by Cyfrin's CodeHawks, analyzing production-grade Solidity codebases for security vulnerabilities.

- **Dating App Audit** — Identified four valid vulnerabilities affecting contract state integrity and authorization logic; submitted comprehensive technical reports with mitigation strategies adopted by project maintainers.

- **Christmas Dinner Audit** — Detected a critical flaw in contract logic that exposed potential re-entrancy and data-validation issues; proposed optimized control-flow patterns and validation checks to enhance protocol security.
- **BriVault Smart Contract Audit** — Identified three high- and medium-impact issues, including DoS attacks, improper reward validation, and faulty incentive logic; recommended stricter state checks and access controls.

Skills & Tools

- **Languages:** Solidity | Python | Node.js | TypeScript
- **Backend:** Node.js | NestJS | ethers.js | The Graph (subgraph indexing) | Cron jobs
- **Version Control & Tools:** Git | GitHub
- **Frameworks & Tooling:** Foundry (forge/cast) | Hardhat | Smart Contract Testing (unit, mainnet-fork)
- **Standards & Patterns:** ERC-20 / ERC-721 / ERC-1155 | EIP-2535 Diamonds | UUPS Proxies | OpenZeppelin
- **DeFi & Protocols:** Uniswap V2/V3/V4 & Universal Router | Permit2 | AMM/CPMM Design | On-chain Order Books
- **Cross-Chain:** LayerZero | Wormhole

Certifications

Blockchain Essentials	IBM	Verify	Apr 2022
Blockchain Basics	Coursera	Verify	Aug 2022
Smart Contracts	Coursera	Verify	Feb 2023
Decentralized Applications	Coursera	Verify	Jul 2023
Uniswap V4	Cyfrin Updraft	Verify	2026

Education

Bachelor of Sciences in Computer Science (BSCS) — Sept 2020 – May 2024
Air University of Aerospace and Aviation Campus Kamra (AUAACK) — **CGPA 3.47 / 4.0**